High school · graphs and logic

Routes without lifting the pencil

Progressive exercises

Exercise 1

A graph has degrees 2,3,3,4. How many edges does it have?

Exercise 2

Prove that the number of odd-degree vertices is even.

Exercise 3

Can a square and both diagonals be drawn without repeating an edge? Interior crossings are not vertices.

Exercise 4

Why do two disjoint cycles not admit one trail even though all degrees are even?